

Project: Loan application process – System analysis

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1. Context

Purpose:

The system allows customers to submit loan applications online. The application is validated, evaluated using a Credit Scoring Service, and then approved or rejected.

Scope of analysis:

Included in MVP:

- Loan application submission
- Customer data validation
- Credit scoring request
- Loan decision

Excluded:

- Payment schedule generation
- Loan contract management
- Customer authentication

2. Process overview

Main workflow:

Customer submits application → System validates data → Credit scoring request → Decision → Approved / Rejected

Actors:

- Customer
- Credit Scoring Service

System:

- Loan Application System

3. Use case

Use case: Submit loan application

Primary actor: Customer

Secondary actor: Credit Scoring Service

Steps:

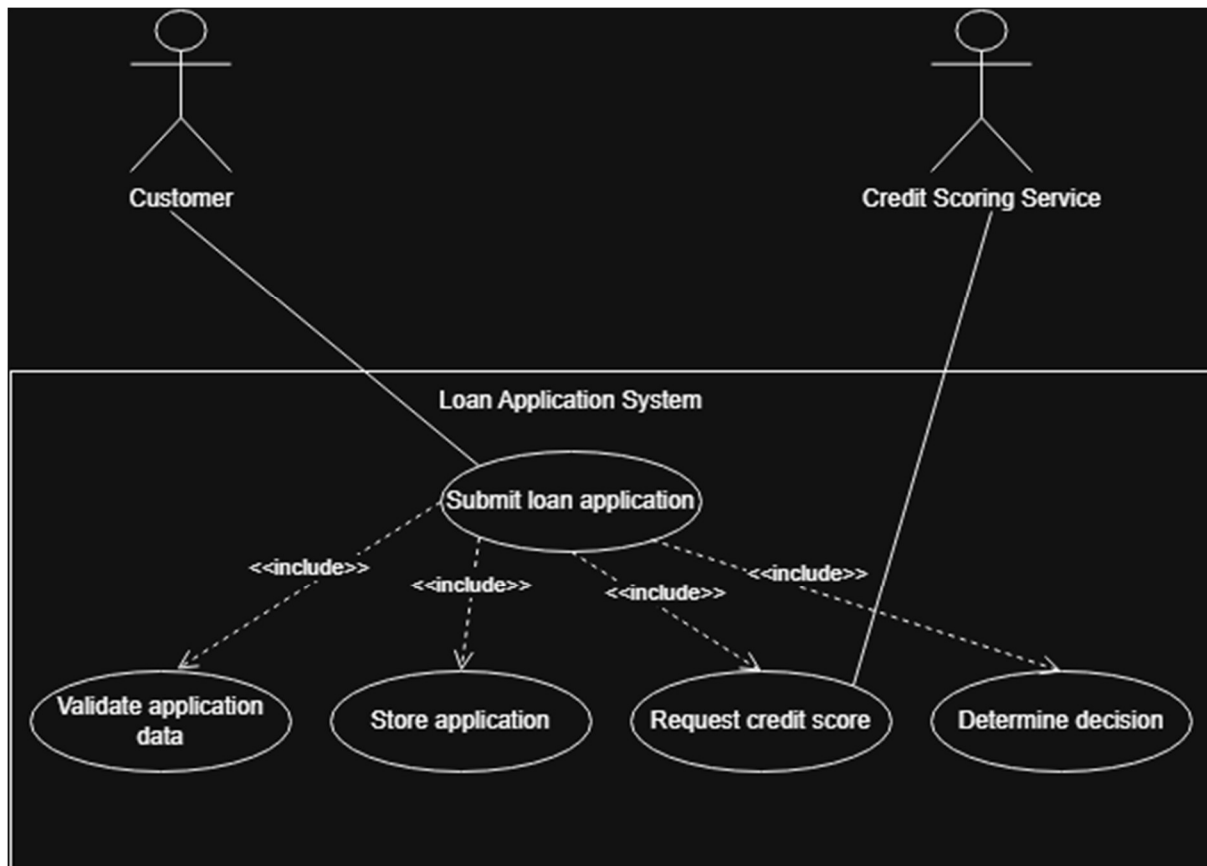
1. Customer fills a loan application form
2. Loan Application System validates required fields
3. Loan Application System stores the application
4. Loan Application System sends a request to Credit Scoring Service
5. Loan Application System receives credit score
6. Loan Application System determines decision

Outcome:

Application status is updated accordingly to:

- Approved
- Rejected

Diagram 1: Use case – Submit loan application



4. Data entities

Customer:

1. A person who submits a loan application
2. Primary key: id
3. Required attributes: name, surname, email
4. Additional attribute: income

Loan_Application:

1. A record of a customer's loan request in the system
2. Primary key: id
3. Foreign key: customerId
4. Required attributes: loanAmount, status
5. Additional attribute: createdDate

Credit_Score:

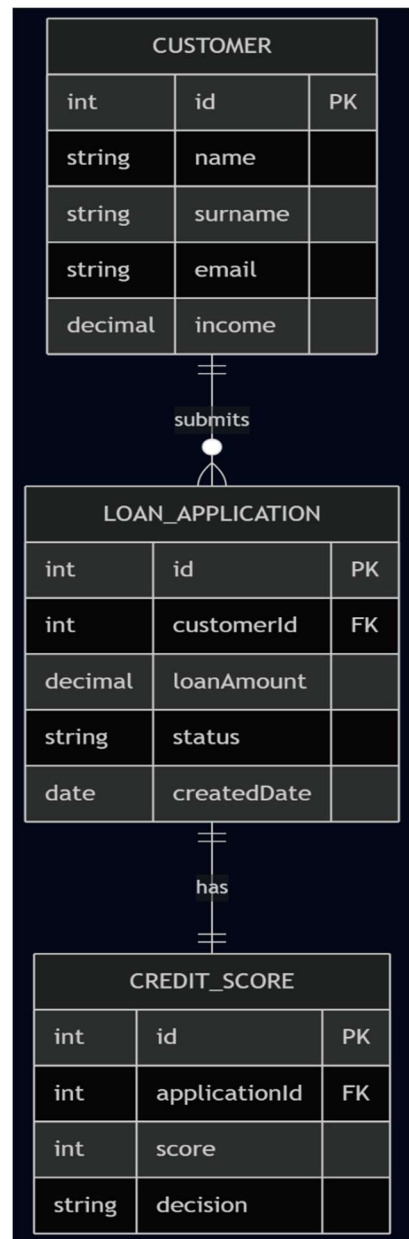
1. A result returned by the credit scoring service for a specific loan application
2. Primary key: id
3. Foreign Key: applicationId
4. Required attribute: score
5. Additional attribute: decision

Relationships:

Customer 1:N Loan_Application

Loan_Application 1:1 Credit_Score

Diagram 2: ERD



5. System interaction

Customer submits a loan application to the Loan Application System.

The Loan Application System validates the submitted data.

The Loan Application System stores the application in the Database.

The Database confirms that the application has been stored.

The Loan Application System sends a credit score request to the Credit Scoring Service.

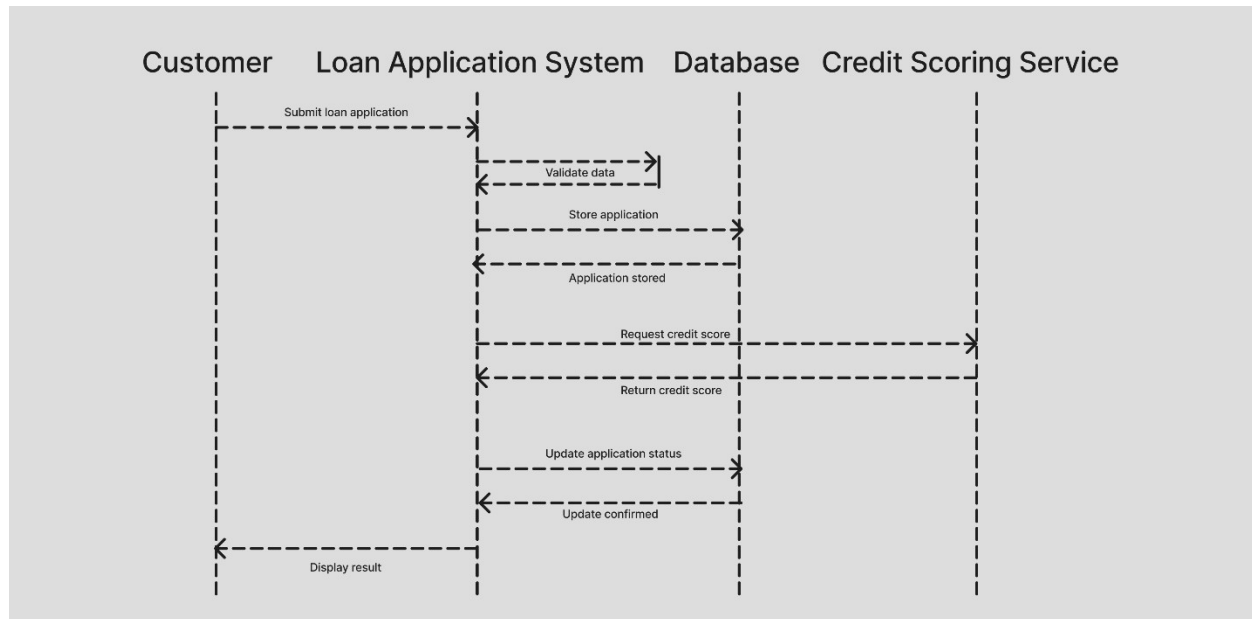
The Credit Scoring Service returns the credit score to the Loan Application System.

The Loan Application System updates the application status in the Database.

The Database confirms the update.

The Loan Application System displays the result to the Customer.

Diagram 3: Sequence – System interaction



6. REST API endpoint suggestions

Customers:

1. Create: **POST** /customers
Req: { "surname": "Novak", "name": "Jan", "email": "novak.jan@gmail.com", "income": 45000 }
Res 201: { "id": 101, "status": "active" }
2. Get customer detail: **GET** /customers/{id}
Res 200 OK / 404 Not Found
3. Edit: **PUT** / **PATCH** /customers/{id}
Res: 200 OK / 400 Bad Request / 404 Not Found

Loan Application:

1. Create: **POST** /applications
Req: { "customerId": 101, "loanAmount": 2000000, "status": "waiting", "createdDate": "2026-03-10" }
Res 201: { "id": 500 }
2. Get application details: **GET** /applications/{id}
Res: 200 OK / 404 Not Found
3. Update: **PUT** / **PATCH** /applications/{id}
Res: 200 OK / 400 Bad Request / 404 Not Found
4. Send application for credit scoring: **POST** /applications/{id}/score
Res: 200 OK / 202 Accepted / 404 Not Found
5. Get final decision: **GET** /applications/{id}/decision
Res 200 OK: { "applicationId": 501, "decision": "Approved" }
Values: Approved / Rejected

7. Validation rules

- loanAmount must be greater than 0
- customer income must be provided
- application cannot be approved without a credit score